# GRIFFIN COLLIER

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Permanent Address: Atlanta, GA
**U.S. Citizen**

## PROFESSIONAL SUMMARY

Mechanical engineering student with 2 years of relevant experience in autonomous aerial systems and counter-UAS technology. Spent a co-op at SCI Technology designing and building AeroGuard, an autonomous counter-drone system, and am now extending that work through aircraft design and flight dynamics research at VSDDL. Experienced in CAD design, composite and additive manufacturing, and fabrication, with working knowledge of electronics, CFD, FEA, and motion analysis. Looking for a full-time aerospace/defense engineering role to keep building autonomous systems.

## EDUCATION

**Auburn University (Auburn, AL) –** Bachelor of Mechanical Engineering, Minor in Finance **Aug 2023 – May 2027**
GPA: 3.77 | Dean's List | Member of Pi Tau Sigma – Mechanical Engineering Honor Society

**Purdue University (West Lafayette, IN) –** Mechanical Engineering coursework (59 credit hours) **Aug 2021 – May 2023**

## APPLICABLE SKILLS

- C, C++, Java, JavaScript, MATLAB, Python
- Siemens NX, SolidWorks (CAD, CAM, PDM & Motion)
- 3D printing (FDM and SLA), slicer software, Ultimaker Cura
- Composites Manufacturing (Carbon fiber layups)
- Manufacturing using waterjet cutting, manual mill and lathe
- Computational Fluid Dynamics (CFD) simulation with FloEFD
- FEA simulation with ANSYS Mechanical, MSC Nastran
- Part 107 Remote Pilot Certified (commercial drone license)

## RELEVANT EXPERIENCE

**AFRL Counter-UAS Challenge, Auburn University – Senior Design Project** **Aug 2026 – current**

- Developing a man-portable counter-sUAS system to protect small tactical teams from Group 1 drone threats for the Air Force Research Laboratory
- Defined a lightweight scouting and surveillance mission with subject-matter experts, and co-authored an 80-page Preliminary Design Review evaluating 3 concepts across the detect, identify, track and neutralize kill chain

**Vehicle Systems, Dynamics, and Design Laboratory (VSDDL), Auburn University – Undergraduate Researcher** **Jun 2026 – current**

- Conducting 15+ hours/week of undergraduate researcher in aircraft sizing and optimization, stability and control analysis, flight simulation, and subscale prototyping of novel aircraft configurations
- Redesigned and built a CNC hot-wire foam cutter for lab fabrication use
- Engineered motion-control electronics and mechanical hardware to increase cutting size and improve reliability
- Designed a custom shipping crate to safely transport a 7-foot subscale research aircraft

**SCI Technology, Huntsville, AL – Mechanical Engineering Co-Op (3 Semesters)** **Jan 2024 – Aug 2025**

- Completed a full year of co-op experience with the same company across three non-consecutive semesters
- Worked full-time to design and build AeroGuard, a fully autonomous counter-drone system that rapidly neutralizes aerial threats
- Designed and manufactured mission critical, aerospace grade hardware
- Gained experience with simulation software including FloEFD for computational fluid dynamics and MSC Nastran for finite element analysis
- Leveraged AI to optimize engineering workflows, improve documentation efficiency, and support project development processes
- Utilized water jetting, SolidWorks, 3D printing, design for manufacturing processes, machining for rapid prototyping, design testing and iteration, and electrical systems pertaining to the defense industry
- Produced professional standard mechanical engineering drawings applying fundamental GD&T principles
- Worked in a multidisciplinary team environment, developing skills in soldering wire harnesses, assembling battery packs, and working with motors, ESCs, and various electrical components to support system integration and functionality
- Performed hardware and flight testing, vehicle analysis, and design iteration to enhance system reliability and optimize efficiency
- Exposed to the Scrum framework within Agile as a project management tool
- Used SolidWorks PDM for engineering data and revision management

**Purdue Aerial Robotics Team, Purdue University – Airframe Team Member** **Aug 2022 – May 2023**

- Designed and manufactured an autonomous UAV for AUVSI-SUAS Competition with focus on surveillance, search and rescue, and payload delivery (Placed 13$^{th}$ overall out of 72 competing teams)
- Spent 200+ hours throughout the year modeling the aircraft in NX, manufacturing MDF carbon fiber molds, performing carbon fiber wet layups for the fuselage and wings, and gluing foam and plywood internal structure
- Gained machining and shop experience with a Dremel tool, drill press, oscillating multi-tool, belt sander, and scroll saw

## ACADEMIC EXPERIENCE

**Office of Career Development and Corporate Relations, Auburn University – Co-Op + Internship Ambassador** **Jan 2025 – Dec 2025**

- Served as a mentor to freshmen and sophomore students to help them navigate the application, onboarding, and job acceptance process
- Delivered presentations to student organizations and classes, sharing my co-op experiences, to promote the Co-Op and Internship Program.
- Collaborated in organizing and managing on-campus career fairs, facilitating employer-student interactions, and supporting corporate engagement initiatives.